

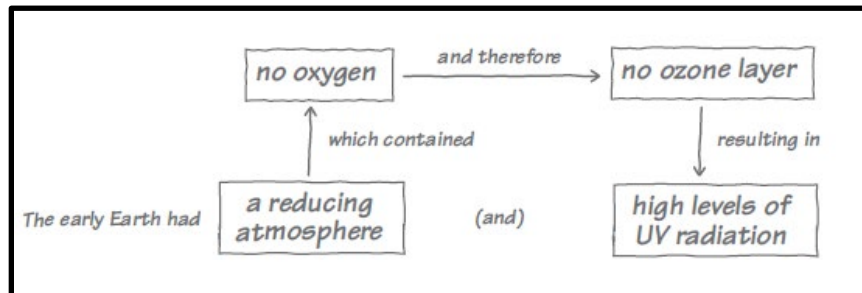
THE HISTORY OF LIFE ON EARTH

Activity 25.1 What do we know about the origin of life on Earth?

Construct a concept map of conditions on the early Earth and the origin of life forms. Be sure to include definitions or descriptions of all the terms in the list below. Keep in mind that there are many ways to construct a concept map.

- Begin by writing each term on a separate sticky note or piece of paper.
- Then organize the terms into a map that indicates how the terms are associated or related.
- Draw lines between terms and add action phrases to the lines that indicate how the terms are related.

Here is an example:



no oxygen

sunlight

ozone layer

DNA

carbon dioxide

high-oxygen atmosphere

amino acids

nucleic acids

organic compounds

prokaryotes

energy source

anaerobic bacteria

antibiotics

reducing atmosphere

electrical discharge (i.e., lightening)

sugars

inorganic compounds

no oxygen

electrical discharge (i.e., lightening)

ozone layer

DNA

carbon dioxide

RNA world

carbon source

cyanobacteria (blue-green algae)

penicillin

high-oxygen atmosphere

amino acids

nucleic acids

organic compounds

reducing atmosphere

sunlight

sugars

inorganic compounds

low levels of UV radiation

Eukarya

mode of nutrition

Gram stain

What type of concept map can you generate using some of the terms above? Sketch below:

Use the understanding you gained from creating the concept map above to answer the questions below.

1. Modern theory suggests that the early (pre-life) atmosphere on Earth was a reducing one. Why (for what reasons) is it believed that oxygen was not present when life formed on Earth?
2. What proposed energy sources existed on this early (pre-life) Earth?
3. In the 1950s, Stanley Miller performed a set of experiments to determine whether life could have evolved given the conditions stated in the answers to questions 1 and 2.
 - a. How was the experiment designed?
 - b. What were the necessary controls?
 - c. What was produced in the experiment?

d. What did the results imply about the possible origin of life on Earth?

e. There is general agreement that life must have evolved in the oceans originally and only much later invaded land. What factors of the physical environment on the early and evolving Earth support these ideas? Changes in which of these were essential to allow life to survive on land?

f. Most of us can't imagine a world without oxygen. However, as you learned earlier, chemically oxygen is a powerful oxidizing compound. What effect(s) would the increase in oxygen levels of the atmosphere have on the organisms that existed at that time?

Activity 25.2 How do we determine the age of fossils and rocks?

To determine the age of fossils and rocks, scientists determine the amounts of radioactive compounds and their stable daughter products present in the sample. Radioactive elements are known to decay into stable daughter compounds at specific rates. A number of radioactive compounds, their stable daughter compounds, and their half-lives are shown in the table below.

Radioactive Compound	Stable daughter compound	Half-life
Carbon 14	Nitrogen 14	5370 years
Potassium 40	Argon 40	1.25 billion years
Rubidium 87	Strontium 87	48.8 billion years
Thorium 232	Lead 208	14 billion years
Uranium 235	Lead 207	704 million years
Uranium 238	Lead 206	4.47 billion years

For dating **rocks**, the potassium-argon method is often used because:

- Argon is a gas. When rock is molten, any existing argon gas will escape. As a result, newly formed rocks contain no argon.
- As the $^{40}\text{potassium}$ naturally occurring in the rock decays to $^{40}\text{argon}$, the $^{40}\text{argon}$ is trapped in pocket in the rock. The ratio of $^{40}\text{potassium}$ to $^{40}\text{argon}$ can be measured to determine the rock's age.

For dating **organic material**, carbon-nitrogen dating is often used because:

- The ratio of radioactive to nonradioactive carbon dioxide in the atmosphere is fairly constant over time. As a result, the levels of these elements in organic tissue remain relatively constant as long as the organism is alive.
- Once the organism is dead, no new inputs of carbon occur and the existing radioactive-carbon-to-nonradioactive-carbon ratio will decrease over time as the radioactive carbon decays.

1. In one half-life, half of the original radioactive compound will decay into its stable daughter compound.

a. If a newly formed rock contains 100 units of $^{40}\text{potassium}$ (^{40}K), how many units of $^{40}\text{pottassium}$ (^{40}K) would it contain after 1.25 billion years?

b. How many units of $^{40}\text{argon}$ (^{40}Ar) would the rock contain when newly formed versus after 1.25 billion years?

c. After the 1.25 billion years, what would be the ratio of ^{40}K to ^{40}Ar in the rock?

d. After 2.5 billion years, what would be the ratio of ^{40}K to ^{40}Ar ?

2. You are fossil hunting and find a trilobite fossil in an old riverbed. You have it radiometrically dated and are told it is 275 million years old.

a. In this amount of time, how many half-lives of ^{40}K have elapsed?

b. Given your answer in part a, what would be the ratio of ^{40}K to ^{40}Ar found in the fossil remains?

3. You want to date some fabric that you have discovered at an archeologic dig.

a. What method of dating would be best for this? Describe the general procedure for dating the cloth.

b. Assume the cloth is 2,000 years old. How would the level of the radioisotope you used to date it have changed to date it have in this period of time?